

AYOUB TOUETI

Junior Computer Science Engineer

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PROFESSIONAL SUMMARY

Junior Software & AI Engineer with hands-on experience in full-stack development, distributed streaming systems, and applied AI. Proficient in Java/Spring Boot, Python, TypeScript/Svelte, Rust, Kafka, Spark, and Docker.

EDUCATION

National School of Engineers of Tunis (ENIT)

Engineering Degree in Computer Science (Major: Software Engineering & AI)

Tunis, Tunisia

Sep 2023 – Sep 2026

Preparatory Institute for Engineering Studies El Manar

Preparatory Cycle in Physics & Technology – National Rank **63/1000** (top 7%)

Tunis, Tunisia

Sep 2021 – Jun 2023

TECHNICAL SKILLS

Core: Java, Python, TypeScript, SQL, Spring Boot, FastAPI, SvelteKit, Kafka, Spark, PostgreSQL, Docker, Git

Also worked with: Rust, PHP, C/C++, React, Angular, Flutter, Node.js, Hadoop, Hive, Cassandra, MongoDB, Redis, RabbitMQ, LangChain, TensorFlow, MinIO, Keycloak, Linux, Firebase

WORK EXPERIENCE

WINGBELS s.r.o.

End-of-Studies Project Intern – AI & Software Engineering

Prague, Czech Republic

Feb 2026 – Aug 2026

- Architected **CyFu**, a multi-agent workflow platform automating ISO 21434 TARA for automotive cybersecurity, modeled as a 13-node DAG with per-step human review gates.
- Engineered an async **FastAPI** backend executing nodes in topological order (Kahn's algorithm) with real-time **SSE** streaming, cutting manual TARA effort by an estimated **60%** across 9 analysis steps.
- Integrated a **LangChain**-based agent service with structured output enforcement and automatic retries, reaching **95%** schema-valid responses across **10+** TARA step types.
- Containerized the full stack (**PostgreSQL**, **Redis**, **MinIO**, **Keycloak**, **TerminusDB**) with Docker Compose and validated it through **100+** automated pytest and API integration tests.

Decclitech

Microservices Developer Intern

Tunisia

2025

- Constructed **3 microservices** for a kids management platform covering child records, enrollment, and administrative workflows, serving **500+** users.
- Modeled **15+ REST endpoints** and relational schemas, reducing average API response time by **20%**.
- Automated integration tests and CI checks, catching roughly **30%** of regressions before deployment.

Tunisie Telecom

PHP Developer Intern

Bizerte, Tunisia

2024

- Created a **PHP/MySQL** fault reporting and navigation tool used by **50+ field technicians**, eliminating manual address entry entirely.
- Embedded **Google Maps** deep links so a technician could jump from a reported fault to turn-by-turn navigation in one click, cutting dispatch time by **40%**.
- Designed the MySQL schema and reporting views for **1,000+** fault records, giving supervisors a live view of pending repairs.

PROJECTS

Kitsune – Desktop Anime Streaming App | Tauri 2, SvelteKit, Svelte 5, TypeScript, Rust

- Launched a cross-platform desktop anime streaming app in **Tauri 2**, **SvelteKit**, **Svelte 5**, **TypeScript**, and **Rust**, integrating AniList/Jikan metadata and Nyaa release search across **1,000+** titles.
- Embedded **librqbit** torrent streaming with stream-health monitoring and external **mpv** handoff, plus OAuth deep-link sign-in and a persistent local watch-progress library.

Web Chat Proxy for Agents | Node.js, Express, Puppeteer, CDP, SSE

- Engineered an OpenAI-compatible proxy routing agent requests to **DeepSeek** and **Qwen** web UIs through **Puppeteer**, exposing a single `/v1/chat/completions` endpoint.
- Captured streaming tokens in real time via **Chrome DevTools Protocol** network interception, achieving **sub-100ms** token forwarding without DOM polling.

Speech Analytics Lambda Architecture | Kafka KRaft, Spark, Cassandra, HDFS, Spring Boot, Avro

- Deployed a Lambda Architecture over the CMU Arctic dataset using **Kafka KRaft**, **Spark Structured Streaming**, **Cassandra**, and **HDFS**, processing **1,132 utterances** at **300 events/sec**.
- Built a speed layer with 10-second micro-batches and a batch layer producing **7 aggregate tables**, surfaced through a Spring Boot WebSocket dashboard.

CERTIFICATIONS & LANGUAGES

Certifications: Cisco CCNA1, Intro to Cybersecurity, Networking Basics (Elementary)

Languages: Arabic (Native), English (B2), French (B2), Italian